6.4.3 Ergodicity of Brownian Dynamics

Imagine a distribution $\rho_0(\mathbf{X})$ which we may take to be an initial macroscopic state of a stochastic differential equation system. This might be a smooth probability density such as a Gaussian, the indicator function ξ_D for a small disk D in the phase space, or, in the extreme case a Dirac delta distribution (indicating that all initial conditions are clustered at a single point in phase space). The density evolves according to the partial differential equation

$$\frac{\partial \rho}{\partial t} = \mathcal{L}^\dagger \rho(\mathbf{X}, t), \quad \rho(\mathbf{X}, 0) = \rho_0(\mathbf{X}),$$

so the solution can be written

$$\rho(z, t) = \exp(t\mathcal{L}^\dagger)\rho_0(\mathbf{X}).$$

Under what conditions (and in what sense) can we say that the distribution obtained in this way converges to a known invariant distribution?

For example, let us return to Brownian dynamics (or overdamped Langevin dynamics) and set $\boldsymbol{M} = \boldsymbol{I}$ and $\gamma = 1$ in (6.36), giving an equation of motion

$$\mathrm{d}\boldsymbol{q} = -\nabla U(\boldsymbol{q})\mathrm{d}t + \sqrt{2\mathrm{k_B T}}\mathrm{d}\mathbf{W}. \tag{6.46}$$

Writing $\beta^{-1} = \mathrm{k_B T}$ as usual, the generator of the stochastic process may be written as

$$\mathcal{L}f = -\nabla U^T \cdot \nabla f + \beta^{-1}\Delta f.$$

We demonstrate a very useful property of this operator.

Let $\bar{\rho}_\beta = \exp(-\beta U)$, and define the space of functions $C_0^2(\bar{\mu}_\beta)$ consisting of functions in C_0^2 on the configuration space $\mathcal{Q} \subset \mathbb{R}^{\mathrm{N_c}}$ together with the inner product

$$\langle f, g\rangle_\beta \stackrel{\text{def}}{=} \int_{\mathcal{Q}} f(\boldsymbol{q})g(\boldsymbol{q})\bar{\mu}_\beta(\mathrm{d}\omega_q) = \int_{\mathcal{Q}} f(\boldsymbol{q})g(\boldsymbol{q})\bar{\rho}_\beta(\boldsymbol{q})\mathrm{d}\omega_q,$$

where $\mathrm{d}\omega_q = \mathrm{d}q_1\mathrm{d}q_2 \ldots \mathrm{d}q_{\mathrm{N_c}}$. Then $\mathcal{L}$ is self-adjoint with respect to this inner product.

The proof is straightforward:

$$\begin{aligned}\langle f, \mathcal{L}g\rangle_\beta &= \int_{\mathcal{Q}} f\left(-\nabla U^T \nabla g + \beta^{-1}\Delta g\right)\bar{\rho}_\beta \mathrm{d}\omega_q \\ &= \int_{\mathcal{Q}} f\left(-\nabla U^T \nabla g\right)\bar{\rho}_\beta \mathrm{d}\omega_q + \beta^{-1}\int_{\mathcal{Q}} f\Delta g\bar{\rho}_\beta \mathrm{d}\omega_q.\end{aligned}$$

Integrating by parts in the second term we have

$$\begin{aligned}\beta^{-1}\int_{\Omega} f\Delta g\bar{\rho}_\beta \mathrm{d}\omega_q &= -\beta^{-1}\int_{\Omega}\nabla g^T\nabla(f\bar{\rho}_\beta)\mathrm{d}\omega_q \\ &= -\beta^{-1}\int_{\Omega}\nabla g^T[\nabla f\bar{\rho}_\beta - \beta\nabla U f\bar{\rho}_\beta]\mathrm{d}\omega_q \\ &= -\beta^{-1}\int_{\Omega}\nabla g^T\nabla f\bar{\rho}_\beta \mathrm{d}\omega_q + \int_{\Omega} f(\nabla g^T\nabla U)\bar{\rho}_\beta \mathrm{d}\omega_q.\end{aligned}$$

Combining this with the first part of the inner product, we see that

$$\langle f, \mathcal{L}g\rangle_\beta = -\beta^{-1}\int_{\Omega}\nabla g^T\nabla f\bar{\rho}_\beta \mathrm{d}\omega_q.$$

In this form we see that f and g appear symmetrically, so we may write

$$\langle f, \mathcal{L}g\rangle_\beta = \langle \mathcal{L}f, g\rangle_\beta,$$

hence, with respect to the weighted inner product, $\mathcal{L}^\dagger = \mathcal{L}$. (Note that the adjoint indicated here is quite different than the L^2 adjoint mentioned previously in connection with the Fokker-Planck operator.)

In addition to being self-adjoint, this operator (on the function space $C_0^2(\bar{\mu}_\beta)$) has other important properties. In particular, it has a *compact resolvent* [360]; this can be shown to imply that the spectrum consists only of a countable collection of eigenvalues, i.e. that the operator has a discrete spectrum. Moreover, from the calculation above,

$$\langle f, \mathcal{L}f\rangle = -\beta^{-1}\int_{\Omega}\|\nabla f\|^2\bar{\rho}_\beta \mathrm{d}\omega_q,$$

so that $\langle f, \mathcal{L}f\rangle \leq 0$ so the operator is negative definite and can only have non-positive eigenvalues. Many additional results are available for this case. A good summary of the state of the art is available in the forthcoming work of G. Pavliotis [298]. In particular, it is possible to show the properties of $\mathcal{L}$ contained in Theorem 6.1.

Theorem 6.1 *Define the generator of Brownian dynamics by*

$$\mathcal{L}f \stackrel{\text{def}}{=} -\boldsymbol{M}^{-1}\nabla U^T\nabla f + \beta^{-1}\Delta_M f.$$